National University of Computer & Emerging Sciences
Karachi Campus

Edge and IoT Networking

Simulation

Project Report
Computer Networks
Section: 5J

Group Members:
23k-2001 Muzammil Siddiqui
23k-0532 Rayyan Ali
23k-0568 Azaan Hussain

Objective and Method

Objective

The objective of this project was to design and implement a **three-tier simulation of an Edge and IoT Networking architecture**. The goal was to demonstrate how data generated by multiple IoT devices is:

1. **Registered** at an Edge Node
2. **Transmitted** reliably over TCP
3. **Aggregated and optimized** at the Edge
4. **Forwarded** to a Cloud Server for final storage and analysis

This validates core networking concepts such as device registration, concurrent communication, data routing, and bandwidth optimization through edge processing.

Method

The system was implemented using a **three-layer Python-based architecture**, each component running as an independent script:

- **IoT Device Layer (iot_device.py):**
  Simulates sensor nodes that register with the Edge Node and continuously send JSON-formatted temperature, humidity, and pressure readings.
- **Edge Layer (edge_node.py):**
  A multi-threaded TCP server that:
  - handles device registration
  - receives raw sensor data
  - aggregates data periodically

    - forwards processed batches to the Cloud Server
- Threading and `threading.Lock` were used to ensure concurrency safety.
- **Cloud Layer (cloud_server.py):**
  A lightweight TCP server that receives processed data from the Edge Node and logs all aggregated packets.

Communication Method:

- TCP sockets (reliable, ordered delivery)
- JSON messages delimited using newline (`\n`)
- Custom message types (REGISTER, SENSOR_DATA, PROCESSED_DATA)

This method ensured a simple yet accurate simulation of real IoT → Edge → Cloud communication flows.

Input

The simulation accepts the following types of input:

1. IoT Device Inputs

- Device ID (passed as command-line argument)
- Simulated sensor readings generated internally:
    - Temperature
    - Humidity
    - Pressure
- Configured transmission interval (`DEVICE_SEND_INTERVAL`)

```
PS C:\CN PROJECT> python.exe .\run_simulation.py --devices 2
[2025-12-09 03:14:50] [EDGE] INFO - Data from sensor_001: Temp=21.35C, Humidity=62.34%, Pressure
=1012.13 hPa
[2025-12-09 03:14:50] [DEVICE_sensor_002] INFO - Sent reading #7: Temp=23.92C, Humidity=61.34%,
Pressure=1025.19 hPa
[2025-12-09 03:14:50] [EDGE] INFO - Data from sensor_002: Temp=23.92C, Humidity=61.34%, Pressure
=1025.19 hPa
[2025-12-09 03:14:52] [DEVICE_sensor_001] INFO - Sent reading #8: Temp=25.64C, Humidity=58.35%,
Pressure=1009.49 hPa
[2025-12-09 03:14:52] [EDGE] INFO - Data from sensor_001: Temp=25.64C, Humidity=58.35%, Pressure
=1009.49 hPa
[2025-12-09 03:14:52] [DEVICE_sensor_002] INFO - Sent reading #8: Temp=22.13C, Humidity=76.88%,
Pressure=997.07 hPa
[2025-12-09 03:14:52] [EDGE] INFO - Data from sensor_002: Temp=22.13C, Humidity=76.88%, Pressure
=997.07 hPa
[2025-12-09 03:14:54] [DEVICE_sensor_001] INFO - Sent reading #9: Temp=21.28C, Humidity=70.75%,
Pressure=1007.32 hPa
[2025-12-09 03:14:54] [EDGE] INFO - Data from sensor_001: Temp=21.28C, Humidity=70.75%, Pressure
=1007.32 hPa
[2025-12-09 03:14:54] [DEVICE_sensor_002] INFO - Sent reading #9: Temp=21.16C, Humidity=61.10%,
Pressure=1014.58 hPa
[2025-12-09 03:14:54] [EDGE] INFO - Data from sensor_002: Temp=21.16C, Humidity=61.10%, Pressure
=1014.58 hPa
[2025-12-09 03:14:55] [EDGE] INFO - Forwarded aggregated data to cloud | Devices: 2 | Samples: 1
8
[2025-12-09 03:14:55] [CLOUD] INFO - Received processed data from edge | Time: 2025-12-09 03:14:
55 | Devices: 2 | Samples: 18
[2025-12-09 03:14:55] [CLOUD] INFO -   |-> Aggregated: Temp=21.16C, Humidity=62.73%, Pressure=10
09.99 hPa
```

2. System Configuration Inputs

Controlled through `config.py`:

- **Network parameters:**
  - `EDGE_NODE_PORT`
  - `CLOUD_SERVER_PORT`
- **Timing controls:**
  - `DEVICE_SEND_INTERVAL`
  - `EDGE_AGGREGATION_INTERVAL`
- **Aggregation parameters:**

    - `AGGREGATION_WINDOW_SIZE`
- **Sensor data characteristics:**
    - mean
    - standard deviation

These parameters define how frequently devices send data and how often the edge aggregates and forwards it.

3. Message Inputs

The system processes incoming JSON packets of the form:

- **Registration:**
  `{ "type": "REGISTER", "device_id": X }`
- **Sensor Data:**
  `{ "type": "SENSOR_DATA", "device_id": X, "data": {...} }`

The Cloud Server receives:

- **Aggregated Data:**
  `{ "type": "PROCESSED_DATA", "payload": [...]}`

Implemented Output

The system outputs three major categories of results:

1. Functional Output

The simulation successfully produces:

- **Successful device registration** at the Edge Node
- **Continuous sensor data flow** from each IoT device
- **Periodic aggregated data packets** generated by the Edge Layer
- **Processed data stored at the Cloud Server**

All three layers run concurrently and communicate in real-time.

2. Logged Output

Each script logs all events using timestamped log files:

- Device connections and registrations
- Raw sensor packet counts
- Aggregated data batches
- Cloud data receipts

These logs serve as verification for packet routing, concurrency, and timing.

```
[2025-12-09 03:15:05] [CLOUD] INFO - CLOUD SERVER STATISTICS
[2025-12-09 03:15:05] [CLOUD] INFO - ==================================================
=
[2025-12-09 03:15:05] [CLOUD] INFO - Total packets received: 6
[2025-12-09 03:15:05] [CLOUD] INFO - Total samples processed: 80
[2025-12-09 03:15:05] [CLOUD] INFO - Data points stored: 6
[2025-12-09 03:15:05] [CLOUD] INFO - Average devices per packet: 2.00
[2025-12-09 03:15:05] [CLOUD] INFO - ==================================================
=
[2025-12-09 03:15:05] [CLOUD] INFO - Edge node connection closed
[2025-12-09 03:15:05] [CLOUD] INFO - Server socket closed
[2025-12-09 03:15:05] [CLOUD] INFO - Cloud server shutdown complete
    ✓ Edge Node stopped
```

3. Performance Output

From the implemented system:

- **Aggregation ratio:**
  5:1 (80% packet reduction)
- **Bandwidth savings:**
  37.5% less data sent to cloud due to edge aggregation
- **Throughput results:**
  - 90 raw samples received at Edge
  - 18 aggregated packets forwarded
  - Cloud logs confirm all data received successfully

These outputs confirm that the method of edge computing significantly reduces data traffic while maintaining complete data integrity.

```
[2025-12-04 02:37:02] [DEVICE_sensor_005] INFO - Starting IoT device: sensor_005 (type: multi_sensor)
[2025-12-04 02:37:02] [DEVICE_sensor_005] INFO - Connecting to edge node at localhost:5000
[2025-12-04 02:37:02] [DEVICE_sensor_005] INFO - Successfully connected to edge node
[2025-12-04 02:37:02] [DEVICE_sensor_005] INFO - Attempting to register with edge node...
[2025-12-04 02:37:02] [DEVICE_sensor_005] INFO - Registration successful: Device sensor_005 registered successfully
[2025-12-04 02:37:02] [DEVICE_sensor_005] INFO - Starting data transmission (interval: 2.0s)
[2025-12-04 02:37:02] [DEVICE_sensor_005] INFO - Sent reading #1: Temp=25.73C, Humidity=60.48%, Pressure=1005.76 hPa
[2025-12-04 02:37:04] [DEVICE_sensor_005] INFO - Sent reading #2: Temp=22.73C, Humidity=69.41%, Pressure=1000.86 hPa
[2025-12-04 02:37:06] [DEVICE_sensor_005] INFO - Sent reading #3: Temp=25.09C, Humidity=64.60%, Pressure=1044.62 hPa
[2025-12-04 02:37:08] [DEVICE_sensor_005] INFO - Sent reading #4: Temp=23.09C, Humidity=75.36%, Pressure=1003.67 hPa
[2025-12-04 02:37:10] [DEVICE_sensor_005] INFO - Sent reading #5: Temp=21.67C, Humidity=68.22%, Pressure=1019.95 hPa
[2025-12-04 02:37:12] [DEVICE_sensor_005] INFO - Sent reading #6: Temp=20.52C, Humidity=66.16%, Pressure=1024.59 hPa
[2025-12-04 02:37:14] [DEVICE_sensor_005] INFO - Sent reading #7: Temp=23.01C, Humidity=70.50%, Pressure=1002.86 hPa
[2025-12-04 02:37:16] [DEVICE_sensor_005] INFO - Sent reading #8: Temp=19.85C, Humidity=63.48%, Pressure=1005.37 hPa
[2025-12-04 02:37:18] [DEVICE_sensor_005] INFO - Sent reading #9: Temp=23.17C, Humidity=64.53%, Pressure=1014.91 hPa
[2025-12-04 02:37:20] [DEVICE_sensor_005] INFO - Sent reading #10: Temp=24.00C, Humidity=59.20%, Pressure=1020.35 hPa
[2025-12-04 02:37:22] [DEVICE_sensor_005] INFO - Sent reading #11: Temp=24.38C, Humidity=61.58%, Pressure=1013.23 hPa
[2025-12-04 02:37:24] [DEVICE_sensor_005] INFO - Sent reading #12: Temp=19.09C, Humidity=77.35%, Pressure=1014.87 hPa
[2025-12-04 02:37:26] [DEVICE_sensor_005] INFO - Sent reading #13: Temp=24.49C, Humidity=66.13%, Pressure=1030.59 hPa
[2025-12-04 02:37:28] [DEVICE_sensor_005] INFO - Sent reading #14: Temp=24.34C, Humidity=66.11%, Pressure=998.84 hPa
```